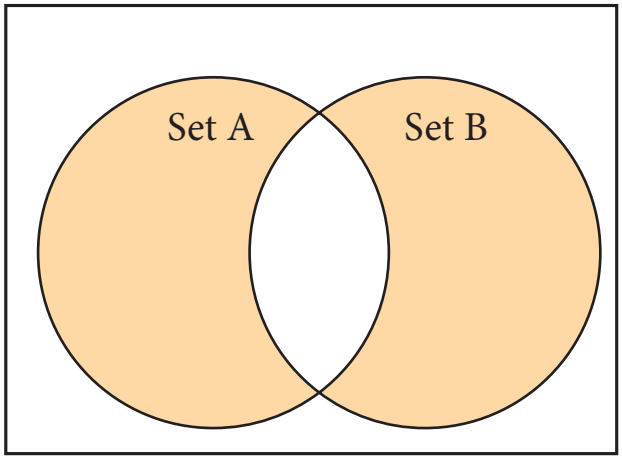
(iv) Symmetric difference

It includes all the elements that are in two sets (say sets A and B) but not the one that are common to two sets.



The caret (^) operator is used to symmetric difference set operation in python. The function **symmetric_difference( )** is also used to do the same operation.

Example: Program to symmetric difference of two sets using caret operator

```
set_A={'A', 2, 4, 'D'}
set_B={'A', 'B', 'C', 'D'}
print(set_A ^ set_B)
```

Output:

```
{2, 4, 'B', 'C'}
```

Example: Program to difference of two sets using symmetric difference function

```
set_A={'A', 2, 4, 'D'}
set_B={'A', 'B', 'C', 'D'}
print(set_A.symmetric_difference(set_B))
```

Output:

```
{2, 4, 'B', 'C'}
```